
Adaptive Inference: Dynamic Efficiency Strategies for Large Language Models

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Abstract

Large Language Models (LLMs) achieve strong performance across many natural language tasks but incur high inference costs, as every input token is processed through all transformer layers regardless of task difficulty Tyukin [2024]. Prior studies show that such uniform computation is often inefficient, since intermediate representations can become sufficiently informative before reaching the final layers Varshney et al. [2023], Skean et al. [2024]. Motivated by this observation, we study adaptive inference using early-exit mechanisms for GPT-2, with the goal of reducing inference latency while preserving predictive accuracy.

We investigate two complementary early-exit strategies: confidence-based exiting and predictive early-exit classifiers. Confidence-based methods terminate inference when intermediate predictions exceed a predefined threshold, while predictive methods attach lightweight multilayer perceptrons (MLPs) to intermediate layers to decide whether to exit. We evaluate these approaches on sentiment classification tasks, where transformer-based models such as RoBERTa are widely regarded as strong industry and academic baselines due to their accuracy and efficiency Rahul et al. [2025], Kanojia and Joshi [2023], Hartmann et al. [2022]. Our results show that adaptive early-exit strategies reduce inference latency by approximately 65–75% while incurring only an 8–9% accuracy reduction relative to RoBERTa, demonstrating their effectiveness for latency- and resource-constrained deployment.

1 Introduction

Large Language Models (LLMs) such as GPT-2 achieve strong performance but require uniform full-depth inference for every input token, leading to high computational cost Wojcik et al. [2023]. Prior work shows that intermediate representations often suffice for accurate classification, enabling early exiting for significant inference speedups. While early exiting has been studied in convolutional networks and encoder-based transformers, its behavior in decoder-only generative models, particularly under fine-tuning, remains less explored Demir and Akbas [2024], Hu et al. [2023].

In this work, we investigate adaptive inference for GPT-2 Medium on sentiment classification using confidence-based and predictive early-exit strategies. We also evaluate shallow GPT-2 classifiers to separate adaptive computation from static depth reduction, and compare all approaches against RoBERTa, demonstrating that adaptive early exiting achieves substantial latency reductions while maintaining competitive accuracy.

2 Background and Related Work

Improving inference efficiency in LLMs has received significant attention due to computational cost and latency constraints on real-world applications. Existing approaches reduce inference cost through model compression, decoding-time acceleration, and adaptive inference, with compression achieved via quantization [Dettmers et al., 2022, Yao et al., 2022, Xiao et al., 2023, Frantar et al., 2022], distillation [Jiao et al., 2020, Mirzadeh et al., 2020, Hsieh and et al., 2023], and structured pruning [Wang and et al., 2020, Guo and et al., 2021]. While effective, these methods typically require specialized training pipelines and result in models with fixed computation budgets.

Speculative decoding accelerates autoregressive generation by using a lightweight draft model to propose candidate tokens that are verified by a larger model [Leviathan et al., 2023]. Such techniques are primarily designed for generative task and are less applicable to single pass classification tasks.

Early exit methods reduce computation by allowing models to terminate dynamically based on input difficulty. For encoder-based models, approaches such as DeeBERT [Xin et al., 2020], PoWER-BERT [Goyal et al., 2020], and DynaBERT [Hou et al., 2020] introduce intermediate classifiers or dynamically adjust model depth. Cascaded systems evaluate inputs through a sequence of increasingly powerful models and exit once sufficient confidence is achieved [Li et al., 2021, Varshney et al., 2022, Yue et al., 2023, Cheng et al., 2023]. For decoder-based models, CALM [Schuster et al., 2022] and depth-adaptive transformers [Elbayad et al., 2020] show that intermediate representations can often support accurate predictions, while LITE [Din et al., 2023] enables intermediate-layer decoding in large decoder-only models, primarily for instruction following and generation.

Our work is conceptually related to prior early-exit approaches but differs in several important aspects. Most existing methods focus on encoder only models or instruction tuned generative models, whereas we study early exiting in *GPT-2* for *classification* task. We demonstrate that a decoder only generative model can act as a strong specialist classifier, achieving competitive performance with RoBERTa while significantly reducing inference latency. In addition to dynamic early exiting, we explicitly evaluate shallow GPT-2 variants, showing that compact models can achieve favorable accuracy & latency trade-offs.

3 Individual Contribution

The project was carried out through close collaboration, with responsibilities divided across model implementation, experimental evaluation, and analysis. While both authors jointly discussed experimental design and results, each contributed primarily to distinct components of the workflow.

Table 1: Individual contributions.

Komal Niraula	Junjie Mai
Repository, dataset preparation, and experiment pipeline management	Literature review and documentation of early-exit and adaptive inference methods
GPT-2-medium model configuration and training setup	RoBERTa baseline setup and evaluation
End-to-end GPT-2 fine-tuning with predictive early-exit classifiers	Zero-shot threshold-based early-exit implementation
MLP-based early-exit classifier design, training, and evaluation	Testing different confidence thresholds on fine-tuned GPT-2
Shallow GPT-2 baseline training and evaluation	Accuracy–latency trade-off evaluation and visualization
Plot generation and contribution to presentation and report	Primary responsibility for presentation slides and report writing

4 Problem Description

LLMs perform uniform computation, requiring every input to be processed through all layers of the model regardless of input difficulty or prediction confidence. This leads to unnecessary inference latency and computational overhead, particularly for classification tasks where many inputs can be resolved using intermediate representations.

At the same time, highly optimized specialist models such as RoBERTa achieve strong classification accuracy with relatively low inference cost, raising an important question: can a general-purpose, decoder-only language model be adapted to act as an efficient specialist without sacrificing much accuracy? Addressing this question is critical for understanding whether generative LLMs can serve as versatile yet efficient models across diverse downstream tasks.

The primary problem addressed in this work is how to reduce the inference latency of a decoder-only language model, specifically GPT-2, for text classification, while preserving competitive predictive accuracy. In particular, we investigate whether adaptive inference mechanisms can exploit the fact that many classification inputs do not require full-depth computation.

5 Approach

Inspired by recent work on enabling intermediate-layer utilization in decoder-only language models, we propose a simple yet effective adaptive inference framework for GPT-2 in the context of text classification.

5.1 Stage 1: Zero-Shot Confidence-Based Early Exiting

We first study early exiting in a zero-shot setting using confidence based criteria. Classification heads are attached to evaluate prediction confidence at a predefined set of intermediate transformer layers.

The model exits at the earliest layer l^* satisfying:

$$l^* = \min \left\{ l \mid \max_y p_l(y \mid x) \geq \tau \right\}, \quad (1)$$

where τ is a confidence threshold.

5.2 Stage 2: Fine-Tuning with Adaptive Early Exiting

We fine-tune GPT-2 on the SST-2 sentiment classification dataset. Each selected intermediate layer is augmented with a classification head, encouraging intermediate representations to become discriminative for the downstream task.

Training minimizes a weighted sum of cross-entropy losses computed at each exit. The overall training objective is

$$\mathcal{L} = \sum_{k=1}^K \lambda_k \mathcal{L}_{\text{CE}}^{(e_k)}, \quad (2)$$

To prioritize early prediction capability, higher weights are assigned to shallower exits, with $\lambda_1 > \lambda_2 > \dots > \lambda_K$. This weighting scheme encourages confident early predictions while maintaining performance at later layers. After fine-tuning, we evaluate two adaptive early-exit strategies.

5.2.1 Confidence-Based Exit Condition

The confidence-based exit strategy for the fine-tuned model follows the same criterion used in the zero-shot stage. Inference terminates at layer l if:

$$\max_y p_l(y \mid x) \geq \tau, \quad (3)$$

where τ controls the accuracy–latency trade-off.

5.2.2 MLP-Based Exit Decision Module

To enable learned exit decisions, we introduce a lightweight MLP-based exit predictor at each exit layer. During training, the GPT-2 backbone and classification heads are frozen, and only the exit predictors are optimized.

At each exit layer l , the input to the exit predictor consists of:

- the hidden state representation h_l from the transformer layer,

- the prediction confidence $\max_y p_l(y \mid x)$,
- the output entropy:

$$H_l = - \sum_y p_l(y \mid x) \log p_l(y \mid x). \quad (4)$$

These features are passed through a one hidden layer MLP with ReLU activation. The exit predictors are trained using a binary cross-entropy loss. The overall training objective is defined as

$$\mathcal{L}_{\text{exit}} = \sum_{l \in \mathcal{L}} \mathcal{L}_{\text{BCE}}(g_l(x), y_l), \quad (5)$$

5.3 Stage 3: Shallow GPT-2 as a Specialist Classifier

In addition to dynamic early exiting, we explore shallow GPT-2 models as efficient specialist classifiers. We truncate the GPT-2 backbone to retain only the first N transformer layers, completely removing deeper layers from computation.

A single linear classification head is attached to the final hidden state of the truncated model. The shallow GPT-2 layers and the classification head are trained jointly in an end-to-end manner using standard cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = - \sum_y y \log p(y \mid x). \quad (6)$$

6 Data and Experimental Setup

6.1 Dataset and Preprocessing

We evaluate our approach on the SST-2 sentiment classification dataset, a widely used binary classification benchmark consisting of positive and negative movie reviews. Each input sample consists of a single sentence paired with a binary sentiment label.

We load the dataset using a unified dataset loader interface, ensuring consistent preprocessing across all experiments. To accommodate variations in dataset formats, we standardize each sample into a (x, y) pair, where x denotes the input text and y denotes the ground-truth class label.

6.2 Verbalizer-Based Classification

Since GPT-2 is a generative language model, we adopt a verbalizer-based classification strategy to map generation outputs to class labels. For SST-2, we define a binary verbalizer as follows:

- negative sentiment: ‘negative’
- positive sentiment: ‘positive’

At each exit layer, the model computes the likelihood of each verbalizer. This approach enables us to use GPT-2 for classification without modifying its vocabulary or generative head.

6.3 Evaluation Protocol

We evaluate all models using a unified evaluation pipeline that measures both classification performance and inference efficiency. To obtain per-sample latency, all predictions are generated in a single-sample inference setting, ensuring that reported latency reflects true end-to-end inference cost per input.

For each evaluation run, we aggregate the following statistics:

- **Accuracy:** the fraction of correctly classified samples.
- **Average latency:** the mean wall-clock inference time per sample, measured in seconds.
- **Average layers used:** the average transformer layer at which inference terminates, serving as a proxy for computational cost.

6.4 Experimental Configuration

Table 2: Experimental setup and evaluation details.

Category	Details
Datasets	Sentiment Analysis (SST-2)
Hardware	NVIDIA A100 GPU
Platform	Google Colab Pro
Metrics	Accuracy (classification performance) Latency (average inference time per sample, seconds) Average number of layers used

7 Results

First, we analyze zero-shot early-exit behavior under different exit-layer groupings to identify effective computation points. Based on this analysis, a representative exit-layer group is selected for fine-tuning. We then compare confidence-based and MLP-based early-exit strategies on the fine-tuned model, evaluate shallow GPT-2 variants, and contrast all approaches against GPT-2 and RoBERTa baselines.

7.1 Zero-Shot Early-Exit Analysis Across Exit-Layer Groups

Across different configurations (Figure 1), we observe a consistent trade-off: lower confidence thresholds lead to earlier exits and reduced latency, while higher thresholds increase latency but do not consistently improve accuracy. Exit-layer groups with densely spaced early layers (e.g., 3, 6, 9, 12, 15, 18, 21) achieve the lowest latency but exhibit substantial accuracy degradation, indicating that early representations are not sufficiently discriminative.

Despite this limitation, we select such configurations for fine-tuning, as they provide the greatest potential for latency reduction and allow us to test whether task-specific training can substantially improve early-exit accuracy while preserving inference efficiency.

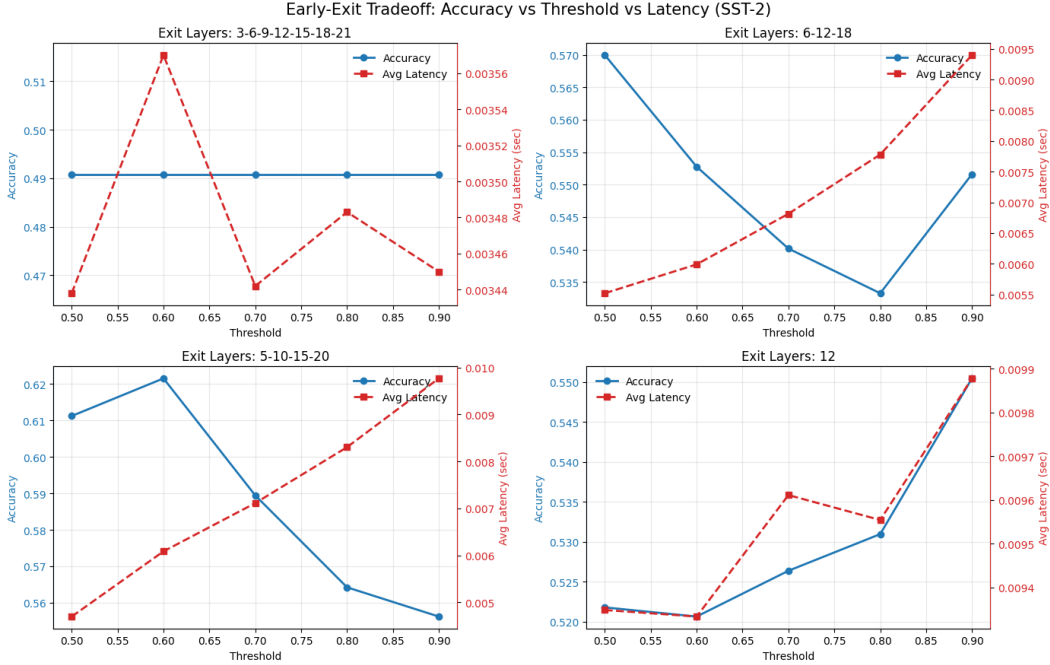


Figure 1: Zero-shot early-exit accuracy–latency trade-offs on SST-2 under different exit-layer group configurations. Each curve is obtained by varying the confidence threshold.

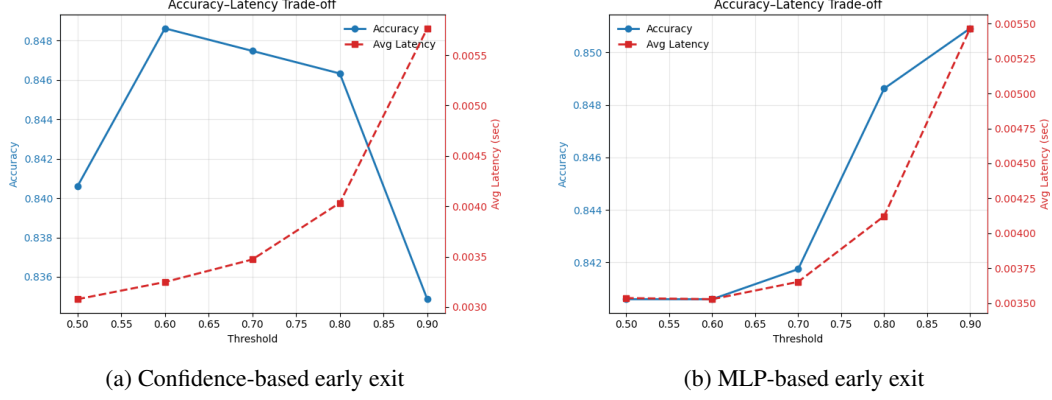


Figure 2: Accuracy latency trade-offs of fine tuned GPT-2 on SST-2 under two adaptive inference strategies: Confidence based and MLP based

7.2 Fine-Tuned Early Exit: Confidence-Based vs MLP-Based

After fine-tuning GPT-2-medium on SST-2 using the selected exit layers, we evaluate two adaptive inference strategies: confidence-based early exiting and MLP-based exit prediction. Figure 2 compares their accuracy–latency trade-offs under different exit thresholds.

The confidence-based approach achieves substantial latency reduction while maintaining accuracy in the range of 83-84% across a wide range of thresholds. However, as the threshold increases, accuracy slightly degrades, indicating that prediction confidence alone is an imperfect proxy for optimal exit decisions. In particular, high-confidence predictions at intermediate layers may still correspond to ambiguous inputs that benefit from deeper computation.

In contrast, the MLP-based exit predictor consistently achieves higher accuracy at comparable latency levels. At a threshold of 0.9, the MLP-based approach reaches 85.1% accuracy with only a modest increase in latency. This improvement suggests that incorporating additional signals, including intermediate hidden-state representations and output entropy, enables the model to make more informed and robust exit decisions.

7.3 Shallow GPT-2 as a Specialist Classifier

We further evaluate shallow GPT-2 variants obtained by truncating the transformer to a fixed number of layers and attaching a task-specific classification head. Figure 3 illustrates the accuracy–latency trade-offs of these fixed-depth models on SST-2.

Compared to full-depth GPT-2, shallow variants substantially reduce inference latency while retaining competitive classification accuracy. Among the evaluated configurations, the 3-layer GPT-2 model achieves the most favorable trade-off, reaching 82.2% accuracy with an average latency of 4.7 ms.

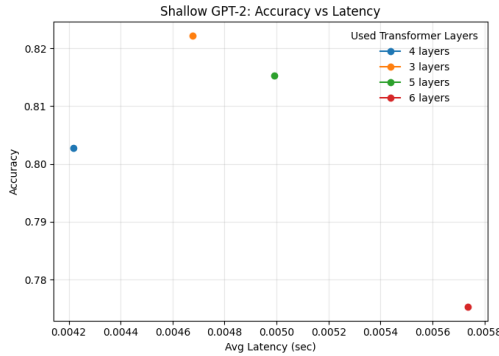


Figure 3: Shallow GPT-2

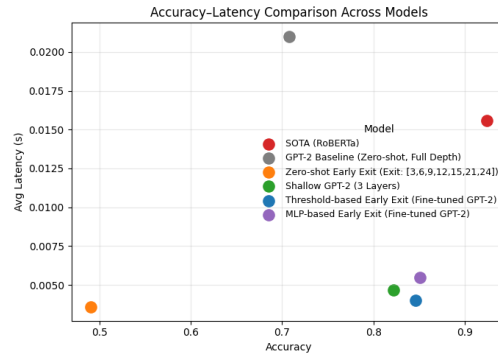


Figure 4: Overall comparison

Table 3: Accuracy and latency comparison relative to RoBERTa

Model	Accuracy	Latency (s)	Accuracy Decrease (%)	Latency Improvement (%)
RoBERTa (SOTA)	0.924	0.01560	0.00	0.00
GPT-2 Baseline (Zero-shot, Full Depth)	0.708	0.02100	23.38	-34.62
Zero-shot Early Exit (Exits: 3,6,9,12,15,21,24)	0.491	0.00357	46.86	77.12
Shallow GPT-2 (3 Layers)	0.822	0.00468	11.04	70.00
Threshold-based Early Exit(Fine-tuned GPT-2)	0.846	0.00403	8.44	74.17
MLP-based Early Exit (Fine-tuned GPT-2)	0.851	0.00547	7.90	64.94

7.4 Overall Comparison with Baselines

Figure 4 presents a comprehensive comparison across all methods in terms of accuracy and inference latency. Zero-shot GPT-2 with full-depth inference performs poorly on SST-2, highlighting the limitations of generative models without task-specific adaptation. Zero-shot early exiting further reduces latency but suffers from severe accuracy degradation.

In contrast, fine tuned early exit methods substantially improve accuracy while maintaining low latency. The MLP based early exit model achieves the best overall trade off among GPT-2 variants, reaching 85.1% accuracy with a 65% latency reduction relative to RoBERTa (Table 3). Although RoBERTa remains the strongest performer in absolute accuracy, our results demonstrate that adaptive inference enables GPT-2 to act as an efficient specialist classifier under strict latency constraints.

8 Conclusion and Future Work

We investigated adaptive inference strategies for GPT-2-medium on text classification, including early exit mechanisms and shallow model variants. Fine tuned early exit models achieve favorable accuracy latency trade-offs, with MLP based exit predictors consistently outperforming confidence based heuristics. Shallow GPT-2-medium models further provide a simple and effective baseline for efficient specialization. Future work can include additional datasets with more challenging classification tasks and extend the approach to more recent language models.

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